AI and Society

Copyright Issues

- Generative AI doesn’t generally copy its training data; learns from it like we do.
- But, can create realistic imitations of actors, voices, and individual musical/artistic styles
- Multiple legal cases ongoing
- What should the rules be?

Liability Issues

- AI makes mistakes, sometimes very damaging
  - Self-driving car that hits pedestrian
  - Incorrect medical diagnoses
- AI cannot be made perfect with present tech
- Who is liable?
  - AI creators, despite impossibility of perfection?
  - AI users, despite fault not lying with them?

AI Quality

- Bias in evaluation and generation
  - Training sets can exhibit bias. Need for “explainability”.
  - Recent efforts by Google Gemini to reverse this produced opposite problem
- Quality issues in generated work
  - Too easy to just accept what the AI generates
- Reduction of human-created training material
  - If AI generates most future content, can it learn from itself?
  - Experiment using LLMs to train LLMs

Concentration of Power

- Concentration of AI in a few companies
  - AI is hugely “compute intensive”
  - Training a modern LLM requires 8 figure expense
  - Training also requires vast libraries of data
- But, the good news..
  - More efficient training and operation is major research area
  - LLMs aren’t everything
  - A culture of knowledge sharing prevails, for now (arxiv.org)

AI in Education

- Should students use AI? Various views:
  - No. It's cheating and they won't learn to create
  - Yes. It's the future; they'll use it all their lives.
  - Students should learn to create w/o AI first, then learn to use it effectively to improve creativity.
- Can AI help teach?
  - Patient tutor
  - Automatic grading, as "first pass", with human final review.

AI and Employment

- Will affect writing, coding, art, law, healthcare, customer service, language translation, entertainment, etc.
- May double or triple productivity of high-level professionals
- My own experience is that it's like working with a widely read and blindingly fast junior engineer, who is also green and makes errors, often subtle ones

Other AI Employment Impact

- It’s not all about LLMs
- Computer Vision
  - CNNs and Vision Transformers process visual data
  - Self-driving cars, inspection tasks, security
- Reinforcement Learning
  - Useful for training AI for manual/robotic tasks
  - Impacts in security, factory production, food service, janitorial, construction

Problems To Consider

- Makes it harder for entry level professionals to learn their trade
- Competition is not between people and AI, but between people who use AI effectively and those who do not.
  - How do we get our students into the first category?
- May result in true technological unemployment
  - Not like going from farm to factory to simple white-collar work
  - “UBI” might provide a living, but doesn’t provide purpose

Case Study – Coding

- CSCI 220 Software Design (Principia College)
  - 2-3x more code completed using AI
  - Much focus on design and reviewing code written by AI
  - Learned new code patterns from AI now and then
  - Also saw subtly wrong code that might have passed testing

Case Study – Art

- Presentations by Apeshka Agarwal
  - Photographer who uses AI extensively
  - https://www.apekshaphotography.com/aiart
  - Makes creative/unusual photos much faster; speeds workflow
  - Edits and fills in
  - Can do scenes not possible with real settings
  - Still “hallucinates” with photographic oddities

Advice to Students

- Learn to Create Without AI First
  - Write, code, do art, etc on your own.
  - Can’t be “in charge” if you are dependent
  - Get closer to content, like doing your own math calculations
- Then, Create Using AI
  - Use AI as a lever to create more, and as a “partner” from whom you can learn
  - This is the workflow of the future
- Focus more on reviewing, editing, large scale design
- Do not complacently accept AI output